Introduction to LaTeX (Cntd..)

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What Are Document and Page Styles?

Editor View:

LaTeX Code

```
% Document class sets the overall style
\documentclass[12pt,a4paper]{article}
% Page style controls headers and footers
\pagestyle{plain}
```

Document view will have:

- The document class (article, report, book, etc.) defines basic structure, font size, and paper size.
- The page style controls how headers and footers appear (plain, empty, headings, fancy).

Choosing a Document Class

Editor View:

LaTeX Code

```
\documentclass[11pt,letterpaper]{report}
```

Document View will have

- Sets document as a report (has chapters, suitable for longer works).
- Font size is 11pt.
- Paper size is US Letter.

Customizing Page Margins

Editor View:

LaTeX Code

```
\usepackage{geometry}
\geometry{left=2.5cm, right=2.5cm, top=2.5cm,
bottom=2.5cm}
```

Document View will have:

- Loads the geometry package to customize page margins.
- Sets all margins to 2.5cm.

Changing Page Styles

Editor View:

LaTeX Code

```
\pagestyle{empty} % No headers or footers
\pagestyle{plain}% Only page number at the bottom
\pagestyle{headings} % chapter name in header
```

Document View will have

- **empty:** No page numbers, headers, or footers.
- **plain:** Only shows page numbers at the bottom.
- **headings:** Section or chapter titles in header.

Advanced Headers and Footers

Editor View:

LaTeX Code

```
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{}
\fancyhead[L]{Left Header}
\fancyhead[C]{Center Header}
\fancyhead[R]{Right Header}
\fancyfoot[C]{\thepage}
```

Document View will have

- Uses the fancyhdr package for full control over headers/footers.
- Customizes text/number on the left, center, and right of header/footer.

Setting Paper Size

Editor View:

LaTeX Code

```
\documentclass[a4paper]{article}
% or for US Letter:
% \documentclass[letterpaper]{article}
```

Document View will have

- Choose paper size by setting options in `\documentclass`.
- Standard sizes: a4paper, letterpaper, etc.

Other Document Classes: Books, Slides, Letters, Exams

- The document class (article, book, report, letter, beamer, etc.) determines the overall structure and behavior of your document.
- Different classes are suited for different types of content.

Editor View:

LaTeX Code

```
\documentclass{article}
```

The Book Class

Editor View:

LaTeX Code

```
\documentclass{book}

\begin{document}
\frontmatter
\chapter{Preface}
\mainmatter
\chapter{Introduction}
\section{Background}
\backmatter
\chapter{References}
\end{document}
```

Explanation:

- Supports chapters, front/back matter, and sections.
- Commonly used for textbooks and dissertations.

The Slides Class (Beamer)

Editor View:

LaTeX Code

```
\documentclass{beamer}

% \begin{document}
% \begin{frame}{Title Slide}
% Welcome to my presentation!
% \end{frame}
% \begin{frame}{Main Points}
% \begin{itemize}
% \item First item
% \item Second item
% \end{itemize}
% \end{frame}
% \end{document}
```

Explanation:

- Provides slide-based layouts for presentations.
- Built-in features for transitions, bullet points, overlays, and more

The Letter Class

Editor View:

LaTeX Code

```
\documentclass{letter}

\signature{Jane Doe}
\address{123 Main St.\\City, Country}
\begin{document}
\begin{letter}{Recipient Name \\ Company \\ Address}
\opening{Dear Recipient,}
This is an example of a letter created in LaTeX.

\closing{Sincerely,}
\end{letter}
\end{document}
```

Explanation:

- Designed for formal correspondence.
- Customizes addresses, openings/closings, and signatures.

The Exam Class

Editor View:

LaTeX Code

```
\documentclass{exam}

\begin{document}
\begin{questions}
\question What is 2 + 2?
\begin{choices}
\choice 3
\choice 4
\choice 5
\end{choices}

\question Explain Newton's second law.
\end{questions}
\end{document}
```

Explanation:

Summary

Class	Use Case	Features
article	Papers, short docs	Sections, no chapters
book	Books, theses	Chapters, parts, front/back matter
beamer	Slides, presentations	Frames (slides), overlays
letter	Correspondence	Address, opening/closing
exam	Tests, quizzes	Questions, choices, solutions

Table: Comparison of Common Document Classes

Creating a Title Page (Standard LaTeX)

Editor View:

LaTeX Code

```
\title{An Introduction to LaTeX}
\author{Alice Example}
\date{\today}

\begin{document}
\maketitle
```

Explanation:

- Displays the document's title, author, and date, centered near the top of the first page.
- Title page formatting varies by document class (book and report classes place the title on a separate page; article on the first page).

Customizing the Title Page

Editor View:

LaTeX Code

```
\title{\Huge \bfseries Custom LaTeX Title}
\author{\Large John Doe \\ \smallskip Department of
    Examples}
\date{August 21, 2025}

\begin{document}
\maketitle
```

Explanation:

- You can use font size and formatting commands (e.g. `\Huge`, `\bfseries`) directly inside the `\title` or `\author`.
- Line breaks and extra space added using `\\` and `\smallskip`.

Adding More Details to the Title Page

Editor View:

LaTeX Code

```
\title{Research on LaTeX Styling}
\author{Jane Author \\
University of Sample}
\date{August 2025}

\begin{document}
\maketitle

% Add additional info below title
\begin{center}
\vspace{1cm}
\textbf{Supervisor:} Dr. Smith \\
\textbf{Course:} Advanced Typesetting
\end{center}
```

Explanation:

Creating an Abstract

Editor View:

LaTeX Code

```
\begin{abstract}
This document introduces the fundamentals of
LaTeX, including document styles, classes, title
pages, and abstract formatting. Readers will
learn to create professional-quality documents
efficiently.
\end{abstract}
```

Explanation:

- The abstract appears as a styled, indented block just after the title page (or the title section), labeled “Abstract”.
- Usually not numbered and formatted differently from the main text

Abstract in Different Classes

Editor View:

LaTeX Code

```
% In book/report/article:
\begin{abstract}
Short summary of the document's purpose and scope.
\end{abstract}

% In Beamer (slides), use a frame:
\begin{frame}{Abstract}
This presentation covers the basics of LaTeX document
    structure, including title pages and abstract
    environments.
%\end{frame}
```

Explanation:

- In most document classes, abstract creates a standard block.
- In Beamer, use a titled frame for an abstract slide.

Customizing the Style:Changing Section Heading Styles

Editor View:

LaTeX Code

```
\usepackage{sectsty}
\sectionfont{\Large\bfseries\color{blue}} % Large, bold,
    blue sections
\subsectionfont{\itshape\color{red}} % Italic red
    subsections

\begin{document}
\section{Introduction}
\subsection{Background}
\end{document}
```

Explanation:

- The `sectsty` package allows changing section fonts globally.
- `\sectionfont` controls the section heading style — here, `\Large`, `\bfseries`, and blue color.
- `\subsectionfont` changes the subsection style to italic and red.

Custom Fonts with fontspec (XeLaTeX/LuaLaTeX)

Editor View:

LaTeX Code

```
\usepackage{fontspec}
\setmainfont{Georgia}
\setsansfont{Arial}
\setmonofont{Courier New}

\begin{document}
Normal text in Georgia.

\textsf{Sans serif text in Arial.}

\texttt{Monospaced text in Courier New.}
\end{document}
```

Explanation:

- The `fontspec` package allows you to use system fonts.
- Use `\setmainfont`, `\setsansfont`, and `\setmonofont` to set serif, sans serif, and monospaced fonts.
- Requires compiling with XeLaTeX or LuaLaTeX.

Changing Page Background Color

Editor View:

LaTeX Code

```
\usepackage{xcolor}
\pagecolor{yellow!10}

\begin{document}
This page has a light yellow background.
\end{document}
```

Explanation:

- The `xcolor` package provides color features.
- `\pagecolor` sets the background color of the entire page.
- The argument `yellow!10` means 10% yellow tint.

Custom Headers and Footers with fancyhdr

Editor View:

LaTeX Code

```
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{}
\fancyhead[L]{My Document Header}
\fancyhead[C]{Center Header}
\fancyhead[R]{\today}
\fancyfoot[C]{Page \thepage}

\begin{document}
Text with customized headers and footers.
\end{document}
```

Explanation:

- The `fancyhdr` package enables flexible header/footer customization.
- `\fancyhf{}` clears existing header/footer content.
- You can set header content for left (L), center (C), and right (R) positions.
- `\thepage` inserts the page number.

Customizing Lists with enumitem

Editor View:

LaTeX Code

```
\usepackage{enumitem}
\setlist[itemize]{label=\textbullet, leftmargin=*}

\begin{document}
\begin{itemize}
  \item First item
  \item Second item
\end{itemize}
\end{document}
```

Explanation:

- The `enumitem` package provides control over list formatting.
- `\setlist` customizes the bullet style and removes indentation with `leftmargin=*`.
- Allows neat, consistent list appearance.

Summary:Customizing the style

Area	**Package/Command**	**Purpose**
Section fonts	sectsty + \sectionfont{}	Font size, color, style
Fonts	fontspec	Use system fonts
Page color	xcolor + \pagecolor{}	Background page color
Headers/footers	fancyhdr + \fancyhead{}	Custom headers and footer
Lists	enumitem + \setlist{}	Bullet style and alignment

Table: Summary of Style Customization Packages and Commands

Manual Line Breaks

Editor View:

LaTeX Code

```
This is the first line.\
This is the second line.
```

Explanation:

- Use \ to break a line manually inside paragraphs.
- This forces the text following \ to start on a new line, but within the same paragraph.
- Do not overuse; LaTeX manages line breaks automatically well.

New Paragraphs and Vertical Space

Editor View:

LaTeX Code

```
This is the first paragraph.

This is the second paragraph.
```

Explanation:

- Leave a blank line to begin a new paragraph.
- New paragraphs have indentation or spacing depending on document style.
- Use `\par` explicitly if needed to end a paragraph.

Page Breaks

Editor View:

LaTeX Code

```
% Force a page break
\newpage

This text starts at the top of the new page.
```

Explanation:

- Use `\newpage` or `\pagebreak` to manually move content to the next page.
- Useful in documents to control page layout for chapters, sections, or figures.
- `\clearpage` also flushes all pending floats (figures/tables) before the break.

Preventing Page Breaks Between Lines

Editor View:

LaTeX Code

```
% Keep words together - no breaks inside box
\mbox{Do not break these words}

% Prevent page break between two paragraphs
\begin{samepage}
Paragraph 1 text.

Paragraph 2 text.
\end{samepage}
```

Explanation:

- `\mbox` keeps its content on the same line without line breaks.
- The `samepage` environment prevents page breaks between enclosed paragraphs.
- Useful for keeping related content together in a tidy layout.

Line-breaking Controls and Hyphenation

Editor View:

LaTeX Code

```
% Suggest allowed hyphenation points
hy-phen-ation

% Disable hyphenation for a word
\mbox{no-hyphenation}

% Allow line break but no page break
\linebreak
```

Explanation:

- Use hyphens inside words to suggest breaking points.
- The `\mbox` disables hyphenation inside its content.
- `\linebreak` can force line breaks with varying strength (optional argument).

Section and Subsection Numbering

Editor View:

LaTeX Code

```
\section{Introduction}
\subsection{Background}
\section{Methods}
```

Explanation:

- Sections and subsections are automatically numbered.
- Numbering hierarchy: Section $\rightarrow$ Subsection $\rightarrow$ Subsubsection.
- The default style shows numbers like 1, 1.1, 1.2, etc.

Controlling Section Numbering

Editor View:

LaTeX Code

```
\setcounter{section}{3} % Start section numbering at 4

\section{Next Section} % This will be Section 4
```

Explanation:

- You can set the starting point for section numbers using `\setcounter`.
- Here, section counter is set to 3, so the next `\section` will be numbered 4.

Suppressing Numbering for Sections

Editor View:

LaTeX Code

```
\section*{Acknowledgments}
```

Explanation:

- Use `\section*` (with a star) to create an unnumbered section.
- Useful for sections like Abstract, Acknowledgments, or Preface.

Equation Numbering

Editor View:

LaTeX Code

```
\begin{equation}
E = mc^2
\end{equation}
```

Explanation:

- The `equation` environment numbers equations automatically.
- Numbers appear on the right by default.
- Use `\label` and `\ref` to reference equation numbers.

Suppressing Equation Numbering

Editor View:

LaTeX Code

```
\begin{equation*}
a^2 + b^2 = c^2
\end{equation*}
```

Explanation:

- Use the starred version `equation*` to suppress numbering.
- Requires `\usepackage{amsmath}` in the preamble.

Customizing Numbering Style

Editor View:

LaTeX Code

```
\renewcommand{\thesection}{\Roman{section}}
\section{Introduction} % Section I
```

Explanation:

- Redefine counters' display format with `\renewcommand`.
- Here, section numbers use Roman numerals instead of Arabic.

Lengths and Spaces in LaTeX

Editor View:

LaTeX Code

```
\setlength{\parindent}{2em} % Paragraph indent
\setlength{\parskip}{1em} % Space between paragraphs

% Horizontal space
This is\hspace{1cm}spaced.

% Vertical space
This is separated by\vspace{0.5cm}vertical space.
```

Explanation:

- `\parindent` controls indentation of paragraphs.
- `\parskip` adds vertical space between paragraphs.
- `\hspace{length}` inserts horizontal space.
- `\vspace{length}` inserts vertical space.

LR Boxes (Save Horizontal Content)

Editor View:

LaTeX Code

```
\newsavebox{\mybox}
\savebox{\mybox}{Some saved content}
\usebox{\mybox}

% Savebox stores content for reuse
```

Explanation:

- LR boxes save horizontally typeset content.
- `\newsavebox` declares the box.
- `\savebox` stores text or elements.
- `\usebox` recalls the stored box anywhere.

Parbox: Paragraph Box

Editor View:

LaTeX Code

```
\parbox{5cm}{
This text is wrapped inside a paragraph box. The width is
    fixed, so the text will break automatically.
}
```

Explanation:

- `\parbox{width}{content}` creates a box with fixed width.
- Content inside behaves like a paragraph, breaking lines as needed.
- Useful for side-by-side text or complex layouts.

Ruleboxes: Draw Horizontal/Vertical Lines

Editor View:

LaTeX Code

```
\rule{3cm}{1pt} % Horizontal rule, 3 cm wide, 1 pt thick
\rule{1pt}{2cm} % Vertical rule, 2 cm high, 1 pt wide
```

Explanation:

- `\rule{width}{height}` draws a filled rectangle.
- Commonly used for underlines, separators, or custom lines.
- Adjust dimensions for horizontal/vertical line thickness and length.

Raising and Lowering Boxes

Editor View:

LaTeX Code

```
Normal text
\raisebox{1ex}{Raised text}
\raisebox{-1ex}{Lowered text}
```

Explanation:

- `\raisebox{distance}{content}` moves content vertically.
- Positive values raise content; negative values lower it.
- Useful for aligning graphics, superscripts, or custom positioning.

Saving and Reusing Boxes

Editor View:

LaTeX Code

```
\newsavebox{\mybox}
\savebox{\mybox}{\parbox{4cm}{This text is saved for reuse
    .}}

% Use saved box multiple times
\usebox{\mybox}
\usebox{\mybox}
```

Explanation:

- Combining `\savebox` with `\parbox` allows complex content storage.
- Reuse saved text or graphics multiple times without retyping.
- Efficient for repeated phrases, logos, or formatted blocks.

Using \framebox to Create Boxes

Editor View:

LaTeX Code

```
\framebox[5cm][c]{Centered text in a box}

\framebox[5cm][l]{Left-aligned text}

\framebox[5cm][r]{Right-aligned text}
```

Explanation:

- `\framebox[width][alignment]{content}` draws a box with specified width.
- Alignment options: `c` (center), `l` (left), `r` (right).
- Text inside is framed, useful for highlighting or layout design.

Using \makebox for Unframed Boxes

Editor View:

LaTeX Code

```
\makebox[4cm][c]{Unframed centered text}

\makebox[4cm][l]{Unframed left-aligned text}
```

Explanation:

- `\makebox[width][alignment]{content}` creates a box without visible border.
- Alignment works as with `\framebox`.
- Useful for spacing or positioning without visible frames.

Using \parbox for Multiline Boxes

Editor View:

LaTeX Code

```
\parbox{5cm}{
This is a paragraph inside a box of fixed width.
The text wraps automatically inside the box.
}
```

Explanation:

- `\parbox{width}{content}` creates fixed-width boxes that can handle paragraphs.
- Line breaks and text wrapping are automatic within the box.
- Great for side-by-side text or complex layouts.

Using \raisebox to Adjust Vertical Position

Editor View:

LaTeX Code

```
Normal text
\raisebox{1ex}{Raised text}
\raisebox{-1ex}{Lowered text}
```

Explanation:

- `\raisebox{distance}{content}` moves content up (positive) or down (negative).
- Useful for aligning different elements visually.

Combining Boxes for Complex Layouts

Editor View:

LaTeX Code

```
\newsavebox{\mybox}
\savebox{\mybox}{
\parbox{4cm}{
\framebox[4cm][c]{Text inside a framed paragraph box}
}
}
\usebox{\mybox}
```

Explanation:

- You can nest boxes, e.g., a `\parbox` inside a `\framebox`.
- Saving the box with `\savebox` allows reuse.
- Useful for reusable complex formatted content.

What Is the Trivlist Environment?

Editor View:

LaTeX Code

```
\begin{trivlist}
\item First item without label
\item Second item
\end{trivlist}
```

Explanation:

- The `trivlist` environment creates a simple vertical list without labels.
- Useful for custom list environments or blocks where bullet/number labels aren't wanted.
- Items inside behave like those in `itemize` or `enumerate` but no markers are shown.

Example: Using Trivlist for Custom Lists

Editor View:

LaTeX Code

```
\begin{trivlist}
\item This text is part of an unlabeled list.
\item You can add paragraphs or other content inside.
\end{trivlist}
```

Explanation:

- Each `\item` adds a new vertical element.
- Unlike `itemize`, no bullet points appear.
- Useful inside custom environments or theorem-like blocks.

Practical Use of Trivlist

Editor View:

LaTeX Code

```
\newenvironment{myenv}
{\begin{trivlist}\item[]}
{\end{trivlist}}

\begin{myenv}
This is a custom environment based on trivlist without item
    markers.
\end{myenv}
```

Explanation:

- You can create custom environments using `trivlist` as the base.
- `\item[]` creates an unmarked item inside.
- This allows precise vertical spacing without visible labels.

Finding and Locating Errors

Explanation:
LaTeX gives errors with line numbers and file info. Use the log to find what happened.
Sample Message:

```
(myfile.tex [10] [11] (part2.tex
! Undefined control sequence.
l.249 \todzy
```

Interpretation: Error in line 249 of `part2.tex`, which is included in `myfile.tex`. The likely cause: typo (should be August 28, 2025).

Error Indicators and Locators

Key Points:

- Error indicator tells what LaTeX thinks went wrong.
- Locator tells where LaTeX was at discovery.
- Remove suspect code and recompile if you can't find the issue.

Common Errors from the PDF

- Bad \line or \vector argument.
- Bad math environment delimiter.
- \begin{...} ended by \end{...}.
- File ... not found.
- Environment ... undefined.
- Missing \begin{document}.
- Lonely \item.
- Option clash for package ...
- Too many columns in eqnarray environment.

Sample Error and How to Fix

Editor View:

```
\begin{tabular}{cc}
A & B & C % <-- Too many columns
\end{tabular}
```

Error: Too many columns in eqnarray/tabular.
Fix: Make sure the number of & in each row matches the column count in format.

Strategy: What to Do If Stuck

1. Check the log and error locations.
2. Undo recent edits or comment out code blocks.
3. Recompile often as you work.
4. Reduce to a minimal document if needed.

TeX Capacity and Overfull/Underfull Warnings

Common resource issues:

- `TeX capacity exceeded`
- `Overfull \hbox/Underfull \vbox`

Solution: Try `\clearpage` or revise text and layout.

MakeIndex: Creating an Index

How to Add an Index:

```
% In preamble:
\usepackage{makeidx}
\makeindex

% In your document:
\index{gnats}
...\printindex
```

Compile: `latex file`, `makeindex file`, `latex file` again.

Types of Index Entries

- **Simple:** `alphalpha00a` $\rightarrow$ alpha
- **Subentry:** `gnat!size ofnat!size of00g` $\rightarrow$ gnat, size of
- **Page range:** `gnus!good|(nus!good|(00g ...`
  `gnus!good|)nus!good|)00g`
- **Cross-ref:** `bat|seebat, atat|seebat, at00b`
- **Custom appearance:** `alpha@\alpha lpha@\alpha 00a` or `alps@alpslps@alps00a`

MakeIndex Special Characters

```
\index{exclaim ("!)} % to include exclamation mark
\index{foo@Bar} % appears as 'Bar', sorted as 'foo'
\index{gnat|emph} % page number in italics for 'gnat'
```

Use " before !, —, or @ to make them normal characters in the index.

Bibliographies with BibTeX

What BibTeX Does:

- Collects references from a `.bib` file.
- Formats the bibliography in various styles.

```
\bibliography{refs}
\bibliographystyle{plain}
```

Sample BibTeX Entry

```
@BOOK{kn:gnus,
  AUTHOR = "Donald E. Knudson",
  TITLE = "1966 World Gnus Almanac",
  PUBLISHER = {Permafrost Press},
  ADDRESS = {Novosibirsk},
  YEAR = 1966
}
```

In the `.bib` file; cite with `\cite{kn:gnus}` in your document.

BibTeX Entry Fields and Types

Type	**Required Fields**
article	author, title, journal, year
book	author/editor, title, publisher, year
inproceedings	author, title, booktitle, year
techreport	author, title, institution, year
misc	none required

Fields like: author, title, year, publisher, journal, volume, etc.

Name and Title Details in BibTeX

- Use last, first or first last: `"Doe, Jane"` or `"Jane Doe"`
- Add 'and' between authors.
- End with 'and others' for 'et al.'
- Protect capitalization: `title = {The {A}gon{y} of {G}nus}`
- Define abbreviations:

```
@string{jgg1 = "Journal of Gnats and Gnus, Series1"}
journal = jgg1
```

- For proceedings/book collections, use `crossref` field to avoid repetition.

Summary of Tips

- Use [fragile] for frames with code or verbatim.
- Only put \documentclass and \usepackage in preamble!
- For indexing, always run latex $\rightarrow$ makeindex $\rightarrow$ latex again.
- For BibTeX, run latex $\rightarrow$ bibtex $\rightarrow$ latex $\rightarrow$ latex.

Thank You!!!!!